

Trade-offs between competition and facilitation: a case study of vegetation management in the interior cedar–hemlock forests of southern British Columbia¹

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Abstract: Vegetation-management practices are applied in temperate-zone forests on the assumption that changing the competitive environment between conifers and unwanted vegetation will improve conifer productivity. We review this assumption using research examining interactions between paper birch (*Betula papyrifera* Marsh.) and conifers in the highly productive Interior Cedar Hemlock zone of British Columbia. We have found that both competition and facilitation are important in young plantations, where paper birch competes for light, reducing growth of shade-intolerant conifers, but having a facilitative effect on shade-tolerant conifers. This facilitative effect may result from greater ectomycorrhizal diversity, population sizes of *Armillaria ostoyae* (Romagn.) Herink antagonistic bacteria, and associative nitrogen fixation in plantations where interior Douglas-fir (*Pseudotsuga menziesii* var. *glauca* (Beissn.) Franco) is mixed with paper birch. Where paper birch is manually cut or girdled, conifers grow faster in diameter, but more die as a result of *A. ostoyae* root disease, and these responses increase with increasing weeding intensity. The weeding treatments do not affect plant community species richness but reduce paper birch dominants and increase understory structural diversity. British Columbia forest policy has been slow to respond to these findings, and we suggest that as a result, the forested landscape incurs substantial risk. We propose additional pathways for managing Interior Cedar Hemlock mixtures to ensure that the natural mix of forest types in the landscape is maintained.

Résumé : Les pratiques d'aménagement de la végétation sont appliquées dans les forêts tempérées en assumant que le fait de modifier le milieu pour réduire la compétition entre les conifères et la végétation indésirable augmentera la productivité des conifères. Nous avons revu cette hypothèse à l'aide de travaux de recherche qui portent sur les interactions entre le bouleau à papier (*Betula papyrifera* Marsh.) et les conifères dans la zone intérieure fortement productive de la Colombie-Britannique caractérisée par la présence de cèdre et de pruche. Nous avons trouvé que la compétition et la facilitation sont aussi importantes dans les jeunes plantations où le bouleau à papier est en compétition pour la lumière, ce qui réduit la croissance des conifères intolérants à l'ombre mais a des effets bénéfiques sur les conifères tolérants à l'ombre. Cet effet bénéfique est possiblement le résultat d'une plus grande diversité des ectomycorhizes, de la taille des populations de bactéries antagonistes vis-à-vis *Armillaria ostoyae* (Romagn.) Herink et de la fixation de l'azote par des organismes associés aux endroits où le douglas vert (*Pseudotsuga menziesii* var. *glauca* (Beissn.) Franco) est mélangé avec le bouleau à papier. Dans les endroits où le bouleau à papier est coupé ou annelé manuellement, les conifères croissent plus rapidement en diamètre mais meurent également plus à cause de la maladie de racines causée par *A. ostoyae* et cela augmente avec l'intensité de l'élimination des espèces indésirables. Les traitements d'élimination des espèces indésirables n'affectent pas la richesse en espèces des communautés végétales mais diminuent le nombre de bouleaux à papier dominants et augmentent la diversité structurelle en sous-étage. La politique forestière de la Colombie-Britannique a été lente à réagir à ces résultats qui, nous croyons, indiquent que le paysage forestier court un risque important. Nous proposons d'autres façons d'aménager le mélange de cèdre et de pruche de l'intérieur pour s'assurer que le mélange naturel de types de forêts soit maintenu dans le paysage.

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Introduction

Broadleaf trees may be both inhibitive and facilitative to the survival and growth of conifers in forest plantations

(Callaway and Walker 1997; Wagner et al. 2005). Inhibition can occur through competition for resources or allelopathy (Harper 1977), whereas facilitation can occur through amelioration of harsh environmental conditions, protection from

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insect or disease-causing pests, or an increase in microbial symbionts (Andersson 1985; Dickie et al. 2005; Lortie and Callaway 2006). Since these interactions can influence plant community succession and ecosystem productivity (Perry 1985), there is considerable interest in understanding the interaction mechanisms and regulating factors affecting the balance between inhibition and facilitation. A number of studies have examined competition and facilitation along resource and stress gradients (Goldberg et al. 1999; Maestre et al. 2005; Lortie and Callaway 2006), but few have quantified the net effect of trade-offs between these interactions within sites. While trade-offs in community ecology commonly involve competition versus colonization, or the differential ability of species to acquire and utilize resources (Kneitel and Chase 2004), the concept can also be applied to competitive and facilitative effects on interactions among plant traits, such as trade-offs between survival and growth (Stearns 1989; Simard et al. 2006).

In managed forests, competition from broadleaf trees is commonly manipulated to increase conifer growth (Wagner et al. 2005), but the possible trade-off with reduced facilitation is routinely ignored. In the northern temperate and boreal forests of western Canada, paper birch (*Betula papyrifera* Marsh.) and trembling aspen (*Populus tremuloides* Michx.) are commonly removed from plantations to allow high-yield conifer production (Lieffers et al. 1996; Comeau et al. 2000; Wagner et al. 2005). This practice is based on the assumptions that broadleaf trees compete intensively with conifers for resources, that their facilitative effects are of less importance to ecosystem function, and that intensive weeding will result in greater conifer productivity (Wagner et al. 2001, 2005). The practice has been supported in several Canadian provinces by the implementation of free-growing policy to regulate competition from broadleaf trees and other vegetation (Brand and Weetman 1986; Comeau et al. 2000; Lieffers et al. 2002). The ecological and economic value of broadleaf trees to forest ecosystems has received increasing attention in Europe and America, however (Swanson et al. 1997; Lautenschlager 2000; Löfman and Kouki 2001; Comeau et al. 2005), raising questions about the effects of a free-growing policy and resultant practices on ecosystem resilience.

Several conifer species commonly associate with broadleaf trees, forming stratified, mixed-species stands in the native temperate-zone forests of western North America (Cameron 1996). Up to 10 species can co-exist on a single site in the Interior Cedar Hemlock (ICH) biogeoclimatic zone of British Columbia (Meidinger and Pojar 1991). For example, shade-intolerant species such as interior Douglas-fir (*Pseudotsuga menziesii* var. *glauca* (Beissn.) Franco), western larch (*Larix occidentalis* Nutt.), lodgepole pine (*Pinus contorta* Dougl. ex Loud. var. *latifolia* Engelm. ex S. Wats.), and western white pine (*Pinus monticola* Dougl.) can regenerate alongside shade-tolerant western redcedar (*Thuja plicata* Donn), western hemlock (*Tsuga heterophylla* (Raf.) Sarg.), interior spruce (*Picea glauca* (Moench) Voss var. *engelmannii* (Parry) Boivin), and subalpine fir (*Abies lasiocarpa* (Hook.) Nutt.), as well as shade-intolerant paper birch, trembling aspen, and black cottonwood (*Populus trichocarpa* Torr. & Gray) following stand-replacing fires (Johnson et al. 1990) or clear-cutting (Simard and Vyse

1994). The density and proportion of regenerating species vary widely depending on seed source, microsite conditions, and degree of disturbance, with localized broadleaf densities ranging between 1000 and 60 000 stems·ha⁻¹ up to 10 years post disturbance (Simard and Vyse 1994). Species variation in height growth and shade tolerance results in early dominance by broadleaf trees and shade-intolerant conifers, but intense vertical stratification quickly gives rise to mixed conifer – broadleaf stands, with eventual dominance by shade-tolerant conifers such as western redcedar and western hemlock. The structural complexity of these forests is maintained through gap-phase dynamics caused in part by the pathogenic fungus *Armillaria ostoyae* (Romagn.) Herink, which causes significant mortality (0.1%–0.2%·year⁻¹ in young stands) among susceptible conifer species. Broadleaf species are resistant to this root disease, allowing them to persist in older forests in gaps created by conifer mortality (Morrison et al. 1991).

Paper birch and trembling aspen can negatively affect conifer growth and survival early in stand development by competing for light and soil resources (Comeau et al. 2003; Simard and Sachs 2004). Paper birch and trembling aspen are pioneer species, however, and their rapid self-thinning, short life-span, and asymptotic height growth pattern (Klinka et al. 2000; Kabzems and Garcia 2004) result in declining competitive effects on conifers as mixed stands develop (Frivold and Frank 2002; Simard et al. 2004b). Free-growing policies have focused on early competition from broadleaf trees for site resources (Brand and Weetman 1986; Lieffers et al. 2002), ignoring the possibility that broadleaf trees can have facilitative effects on conifer regeneration, and that competitive effects can vary over time and space. It has previously been shown, for example, that broadleaf trees help limit the spread of root disease among conifers (Morrison et al. 1988; Gerlach et al. 1997), reduce the risk of conifer attack by weevils and spruce budworm (Taylor et al. 1994; McIntosh et al. 1996), protect understory conifers against frost (Andersson 1985; Pritchard and Comeau 2004), provide habitat for ungulates, small mammals, and birds (Peterson et al. 1997; Aitken et al. 2002), and contribute to soil productivity through nutrient-rich litter inputs (Brockley and Sanborn 2003). In keeping with this last point, total yield (conifers plus broadleaf trees) in mixed forests has sometimes been higher than in pure coniferous forests because of niche separation (Kenk 1992; Mielikainen 1996; Frivold and Kolström 1999; Man and Lieffers 1999; Valkonen and Valsta 2001).

The typical management pathway for achieving free-growing high-yield conifer stands in temperate-zone mixed forests over the last 20 years can be described as intensive, and is aimed at increasing conifer survival and growth rates through reduction in competition from broadleaf trees (Table 1). In British Columbia, standards governing free-growing plantations were recently modified (British Columbia Ministry of Forests 2000) to allow a conifer two to five neighboring broadleaf trees (the number depending on the conifer species) within a 3.99-m radius, provided that they are shorter than the conifer in three of four quadrants (British Columbia Ministry of Forests 2000). Broadleaf trees can be included as a component of acceptable stands only where the effect on crop-tree growth is minimal. This condition is

Table 1. The current management pathway and three alternative pathways for managing mixtures across the landscape.

Management pathway	Management action				Management aim	
	Site preparation	Stumping for root-disease management	Regeneration	Vegetation control	Stand	Landscape
Current pathway: conifer sawlog production	Mechanical	Yes	Planting at moderate density; two or three early-successional conifer species	Manual brushing to meet free-growing regulation	Mixed conifer and minor broadleaf trees, even-aged, single storey	Clearcut
First pathway: conifer enrichment	Mechanical or none	None	Planting two or three early- and late-successional conifer species at low density	Selective manual brushing	Stratified mixture of conifers and broadleaf trees	Continuous cover
Second pathway: semi-natural management	None	None	No planting	No brushing	Uncontrolled stratified mixture of conifers and broadleaf trees	Continuous cover
Third pathway: high-intensity management	Mechanical	Yes	Planting single early-successional species at moderate to high density	Broadcast chemical brushing	Single species, conifer or broadleaf trees, even-aged, single storey	Clearcut

rarely encountered because trembling aspen and paper birch grow more rapidly than most conifers. Thus, almost none of the conifer plantations in the interior cedar–hemlock forests meet the requirements for free-growing plantations without a brushing treatment to remove the broadleaf trees. The entire management pathway for achieving free-growing plantations is as follows. Following harvesting of root-disease-infested sites, infected stumps are usually extracted to reduce rhizomorph production and inoculum loads of *A. ostoyae* (Morrison et al. 1991). This is rapidly followed by planting of large container-stock conifer seedlings of one or two early-succession species, usually Douglas-fir or lodgepole pine. These species are chosen because of their current commercial value, rapid juvenile height growth allowing early attainment of free-growing status, and broad edaphic amplitude (British Columbia Ministry of Forests 2000; Klinka et al. 2000). Prior to the assessment of free-growing status, the plantations are weeded of broadleaf trees and other encroaching shrub species, using herbicides or manual cutting. Spacing to reduce conifer densities is commonly applied either with the weeding operation or a few years later, favoring the early-succession conifer species. Even with the recent modifications allowing some broadleaf trees, the general intent remains one of producing early seral conifer sawlogs. Based on international standards this management pathway is expensive (Simard and Vyse 1994), costing in 2005 up to \$3000/ha where repeated manual treatments were necessary. It follows the basic principles of management for high-yield conifer production practiced elsewhere in the world (Wagner et al. 2005).

Extensive application of similar intensive-management pathways is raising concerns about the diversity, health, and resilience of western North American forests (Lieffers et al. 2002; Woods et al. 2005). In addition to their lower structural diversity (Haeussler and Bergeron 2004), these conifer plantations have been subject to widespread insect

and disease damage, such as mountain pine beetle (*Dendroctonus ponderosae* Hopkins), Dothistroma needle blight (*Dothistroma septosporum* (Dorog.) Morelet), and *A. ostoyae* root disease (Cruickshank et al. 1997; Woods et al. 2005). This kind of damage, along with mortality caused by drought and fire, is expected to increase in southern interior British Columbia with changing climatic conditions (Hansen et al. 2005; Hamann and Wang 2005). Not only is this management pathway expensive, but it may be placing our forests at an elevated risk of future problems in the face of climatic uncertainty.

The objectives of this article are to review our understanding of interspecific interactions between conifers and broadleaf trees in the mixed temperate-zone forests of interior British Columbia, and to examine whether management practices aimed at achieving free-growing conifer plantations result in a trade-off between the competitive versus the facilitative effects of broadleaf trees on conifers. Most of the review is focused on our own work because it constitutes a substantial portion of the research conducted in these southern interior forests, but we support and complement it with the results of work done elsewhere in similar forest types. We follow this review with a comparison of management pathways that may be applied across the landscape, ranging from a low-intensity “do nothing” approach to a more intensive management pathway than the current management practices described above. We end with a discussion of new directions for policy and management.

Interspecific interactions

Our research shows that broadleaf trees are both facilitative and competitive to conifer establishment and growth in mixed forests, and that the relative strength of these interactions varies over time and space and with conifer species (Simard et al. 1997a, 2004b, 2005; DeLong et al.

2002; Simard and Sachs 2004; Baleshta et al. 2005). During seedling establishment (to 3 years old), soil rhizosphere microbes associated with paper birch facilitate ectomycorrhizal carbon transfer, associative nitrogen fixation, and *A. ostoyae* antagonism, potentially favoring Douglas-fir survival and growth of seedlings. When the young trees are saplings (3–15 years old), rapidly growing paper birch overtop Douglas-fir, and can reduce light and water availability as well as growth rates of the conifers. Once the stands reach crown closure (at 15–20 years), however, the competitive effects of paper birch diminish with increasing conifer dominance and shading. By 50 years, paper birch has substantially self-thinned from the mixed stands, and inter-conifer competition for soil resources is a much stronger determinant of conifer growth rates than broadleaf competition. Over the long term, ecosystem modeling suggests that maintaining a component of paper birch is necessary for maintaining productivity of Douglas-fir, owing to its nitrogen inputs (Sachs 1996).

Stand-establishment stage

When stands are establishing, paper birch and conifer root systems are quickly inoculated with mycorrhizal fungi, with greater evenness (Jones et al. 1997), richness (Simard et al. 1997a), and diversity (Simard et al. 1997c) of fungal types on Douglas-fir when it is grown in mixture with paper birch than when it is grown alone. We suggest that this tree species “mixture effect” on the ectomycorrhizal fungi is beneficial to Douglas-fir because mycorrhizal fungi differ in their abilities to break down organic nutrients, transport water, protect against pathogens, and colonize different types of soil, thus conferring increased physiological diversity on the Douglas-fir seedlings. By the time seedlings are 1 year old, more than half of the root tips of paper birch and Douglas-fir are inoculated with the same fungal types (Jones et al. 1997; Simard et al. 1997a), potentially linking the tree species in a common mycorrhizal network (CMN; Simard and Durall 2004). We found that carbon can transfer between paper birch and Douglas-fir through this CMN, with net transfer from birch to fir when seedlings are 2 years old (Simard et al. 1997b). We suggest that carbon is transferred through the CMN in combination with nitrogen as amino acids along a source–sink gradient, with paper birch functioning as a source and Douglas-fir as a sink for carbon and nitrogen. At this age, net photosynthetic rates, fine root carbon allocation, and tissue nitrogen contents of paper birch seedlings are more than double those of Douglas-fir (Wang et al. 1996; Simard et al. 1997b), which agrees with broadleaf conifer comparisons in many other studies (Perry 1994). The high nitrogen content of paper birch may result from associative N_2 fixation by the rhizosphere bacteria, with rates of fixation in these forests (estimated up to $25 \text{ kg} \cdot \text{ha}^{-1} \cdot \text{year}^{-1}$) being higher among fine than among coarse paper birch roots (Blenner-Hassett 1996; Simard 1996). Paper birch grows taller than Douglas-fir at this early age, but the more Douglas-fir is shaded by paper birch, the more carbon (and potentially nitrogen) it receives from paper birch through the CMN, representing approximately 10% of total carbon fixation (Simard et al. 1997b). We found that 2- and 3-year-old Douglas-fir linked into a CMN with either same-age or mature (80 years old) paper birch grew significantly taller and had higher foliar nitrogen concentrations than those that

were isolated from paper birch by means of trenching (Simard et al. 1997c; Baleshta 1998).

When conifer seedlings in this forest type are regenerating following harvest, they can also suffer significant levels of *A. ostoyae* infection and mortality when contacting inoculated root systems remaining in the soil (Morrison et al. 1991). Any subsequent silviculture treatment that increases the pathogen inoculum load, such as brushing, spacing, or thinning, can cause additional mortality among susceptible seedlings (Cruickshank et al. 1997; Morrison et al. 2001; Simard et al. 2005). While root extraction, or stumping, has been the treatment of choice for mitigating *A. ostoyae* root disease in young conifer plantations in British Columbia (Morrison et al. 1991), retaining resistant paper birch in intimate mixture also appears to reduce disease incidence among the neighboring susceptible conifers (Morrison et al. 1988). Several mechanisms have been proposed to explain this reduced conifer mortality, including reduced probability of attack, discontinuity of available inoculum, and birch-induced changes in soil microbiology. We have found evidence for the latter — young paper birch stands contained a population of *Pseudomonas fluorescens* Migula, 1895 which was 4 times that of young Douglas-fir stands — and that most isolates reduced growth of *A. ostoyae* in dual culture tests (DeLong et al. 2002). Paper birch appears to provide a more favorable environment for *P. fluorescens* than Douglas-fir, which suggests a mechanism by which paper birch positively influences the susceptibility of regenerating conifers to *A. ostoyae* root disease (DeLong et al. 2002).

Sapling stage

Once healthy seedlings are established and growing, paper birch gains a height advantage over Douglas-fir, as well as all other neighboring conifer species, except for rapidly growing individuals of western larch or lodgepole pine. Three years after establishment, we found that height growth and net photosynthetic rates of Douglas-fir in root contact with paper birch (untrenched; Baleshta 1998) lagged behind those that were isolated from paper birch (trenched), suggesting that the competitive effects of paper birch on Douglas-fir had become more important than its facilitative effects. At this age, the taller paper birch compete with Douglas-fir for available light, but also have larger root systems, which reduce soil water content, as is shown in the study of an untrenched compared with a trenched treatment by Baleshta (1998) (see also Simard and Durall 2004). By the time single-cohort plantations are 10 years old, overtopping paper birch trees strongly attenuate light through their canopies, with transmittance (photosynthetic photon flux density (PPFD) as a percentage of full sunlight) at 1 m height ranging from as low as 10% with 10 000 overtopping stems·ha⁻¹ to 60% with 1500 stems·ha⁻¹ and almost 100% in the absence of broadleaf trees (Simard and Sachs 2004). Conversely, Baleshta et al. (2005) found that reducing broadleaf density increases %PPFD from 40% at 7000 broadleaf stems·ha⁻¹ to 62% at 1111 stems·ha⁻¹, 70% at 400 stems·ha⁻¹, and 82% at 0 stems·ha⁻¹ (not 100% because of shading by neighboring conifers). Baleshta et al. (2005) also found that soil water content in August increased linearly with declining broadleaf density (from 9% to 15% with complete removal of broadleaf trees), suggesting that

paper birch competes with Douglas-fir for soil water. The light-attenuation effects measured by Simard and Sachs (2004) and Baleshta et al. (2005) agree with Comeau et al. (1998), who found that growing-season PPFD declined logarithmically with increasing paper birch density in 35-year-old stands.

Light availability strongly affects understory conifer growth rates, and several studies show that the magnitude of this effect varies with conifer shade tolerance (Wright et al. 1998; Drever and Lertzman 2001). Shade-tolerant western redcedar, western hemlock, and subalpine fir, for example, reach 50% of their maximum height growth at 25% full sunlight (Wright et al. 1998), while shade-intolerant Douglas-fir and lodgepole pine reach 60%–90% of their maximum at 60% light (Wright et al. 1998; Drever and Lertzman 2001). In keeping with this, Simard and Sachs (2004) found that in 10-year-old mixed plantations, diameter growth of shade-intolerant western larch that had become overtopped by broadleaf trees was reduced more severely with light attenuation than was shade-tolerant western redcedar. Diameter growth of western redcedar was actually positively correlated with abundance of broadleaf neighbors (Simard et al. 2004b), suggesting a prolonged facilitative effect of paper birch, possibly because turnover of nutrient-rich litter helped to meet the redcedar's high nutrient requirements (Klinka et al. 2000). In spite of these effects, we have found that light level is a poor predictor of diameter growth rates of conifers in young ICH stands, where the vertical and horizontal distribution of broadleaf trees is highly variable, with most conifers growing between paper birch clumps and interspecific interactions thus confounded by variable degrees of crowding as well as root disease. Light availability under 10-year-old paper birch canopies accounted for only 19%–38% of the variation in diameter of same-age Douglas-fir (Simard et al. 2004b), contrasting with more predictable effects of high overstory on understory saplings in older northern sub-boreal and temperate-zone forests (Wright et al. 1998; Comeau et al. 2003).

Stem-exclusion stage

With canopy closure at 15–25 years, prior to the onset of significant self-thinning, the importance and intensity of inter-tree competition reach an asymptote in these forests (Simard et al. 2004b). At this time, individual tree growth becomes relatively more affected by soil-resource (water and nutrient) competition from shorter neighbors than by light competition from taller neighbors (Simard and Sachs 2004). Because of its high water and nutrient demands, and more suppressed position in the canopy, western redcedar is more strongly affected by competition from neighbors in these older stands than are Douglas-fir and western larch (Simard and Sachs 2004). Broadleaf trees are still important competitors of conifers at 25 years, but their declining relative height growth rates place them at a comparative height disadvantage with respect to conifers by 50 years (Wang and Kimmins 2002). By 50 years, broadleaf trees have thinned considerably from the stands, and competition from other conifers for soil resources is of greater importance than competition from broadleaf trees (Simard et al. 2004b). Most broadleaf trees have fallen out of the forests by 70 years because of overtopping by the longer lived conifers

(Wang and Kimmins 2002). At this time, broadleaf trees persist only where gaps created by root disease, insect mortality, or windthrow are large enough to meet their light requirements (Simard and Vyse 1994).

Long-term productivity

Ecosystem modeling suggests that nitrogen inputs by paper birch are important for maintaining the productivity of mixed conifer–broadleaf forests over several rotations. Sachs (1996) calibrated the FORECAST ecosystem model (Kimmins et al. 1990) using growth and nutrient information collected in mixed paper birch – Douglas-fir stands in ICH ecosystems (e.g., Wang et al. 1996). Mixedwood-management scenarios over four 100-year rotations (400 years) were simulated using a replacement series design with a constant density of 1600 stems·ha⁻¹ (Harper 1977). Sachs (1996) predicted that Douglas-fir productivity was lower than that of paper birch, declining steadily and significantly over 400 years when Douglas-fir was grown in pure stands. However, Douglas-fir yield increased and then stabilized over eight rotations with inclusion of at least 400 stems·ha⁻¹ paper birch in mixture. The model predicted that total yield would be higher in mixed stands of 400 stem·ha⁻¹ paper birch and 1200 stems·ha⁻¹ Douglas-fir than in pure stands of either species. This study suggests that paper birch, because of its high tissue nitrogen content and associative nitrogen fixation by rhizosphere bacteria, is important in maintaining long-term productivity of ICH forests (Blenner-Hassett 1996; Sachs 1996).

Weeding effects on conifer growth and survival

Competition thresholds

High densities of paper birch can negatively affect growth of conifers in young stands, but birch density is spatially variable, with high densities occurring in small patches and most mesic microsites supporting low to moderate birch densities that have small and variable effects on conifer growth (Simard et al. 2004b). In 10-year-old Douglas-fir plantations, Simard (1990) examined conifer growth across a range of neighbor birch densities and found that 340–2100 stems·ha⁻¹ of 3 m tall paper birch had little effect on conifer diameter growth. In a later study, Simard et al. (2001) found that diameter growth of 10- to 15-year-old Douglas-fir was highly variable at low densities of 2–3 m tall paper birch, and was consistently suppressed only at densities above 4400 total birch stems·ha⁻¹ or 2500 taller birch stems·ha⁻¹. In these studies, high densities of paper birch were much less common than low densities. Current free-growing regulations in British Columbia allow lower retention of broadleaf trees than was suggested by Simard (1990) and Simard et al. (2001). In the following section we describe the results of several experiments in which we tested these competition thresholds by manipulating birch densities using a variety of weeding methods.

Growth

In several experiments we have found that diameter growth of young conifers generally increases with intensity of broadleaf reduction, whereas height is unaffected.

Baleshta et al. (2005) found that 10- to 15-year-old Douglas-fir increased in mean diameter increment when broadleaf trees were completely removed or heavily thinned to 400 stems·ha⁻¹, but not where they were thinned to moderate density (1111 stems·ha⁻¹), suggesting that broadleaf densities above 400 stems·ha⁻¹ suppressed Douglas-fir growth potential. In a separate mixed plantation, Simard and Hannam (2000) similarly found that reductions of 8-year-old paper birch from 2500 to 50 stems·ha⁻¹ significantly improved diameter growth of interior spruce. Simard et al. (2005) tested the effect of broadcast weeding over 20 sites using three methods and found that Douglas-fir diameter increased by 37% and 17% following the cut-stump glyphosate and girdling treatments, respectively, when broadleaf density was reduced below 4400 stems·ha⁻¹, or 15% cover. By contrast, it was unaffected by manual cutting, where most broadleaf trees vigorously sprouted back to an average density of 10 812 stems·ha⁻¹, or 25% cover. As predicted by the competition threshold model of Wagner (2000), these studies together show that the greatest average conifer-diameter response occurred where broadleaf trees were reduced to low levels (below 5% of the maximum broadleaf density on the sites), and small responses started at intermediate levels (approximately 20% of the maximum broadleaf density). Our results fall within the competition thresholds of Simard (1990) and Simard et al. (2001), where maximum responses between 0 and 400 stems·ha⁻¹ and minimum responses between 1000 and 4400 stems·ha⁻¹, or ≤15% cover, were predicted (Simard and Heineman 1996; Baleshta et al. 2005; Simard and Hannam 2000). They also agree with the majority of competition studies, which show maximal growth gains between 5% and 10% cover and minimal gains starting below 20% (Wagner 2000). In none of these studies have we found evidence of a competition threshold for height losses, reflecting the relative insensitivity of height to competition (Wagner and Radosevich 1998).

Differences in conifer species responses to weeding reflect earlier predictions based on shade-tolerance rankings (Simard and Sachs 2004). Simard et al. (2005) found that lodgepole pine, for example, responded to smaller decreases in paper birch density than interior Douglas-fir. Lodgepole pine is a relatively shade-intolerant species and highly plastic in its response to increasing light availability (Wright et al. 1998), allowing it to respond more readily to moderate increases in light than Douglas-fir. Lodgepole pine also has a more rapid juvenile height growth rate (Klinka et al. 2000), putting it in a better competitive position with respect to broadleaf trees. Other studies have also shown that shade-intolerant species with fast juvenile growth respond more dramatically to release from competition with broadleaf trees than shade-tolerant species do (Frivold and Frank 2002).

Survival

In none of these studies did reduction of broadleaf density improve conifer survival. This pattern, where conifer survival is restricted at higher competition levels than growth, agrees with the competition-threshold model of Wagner (2000). In stark contrast to this model, however, mortality has consistently increased following manual weeding of broadleaf trees because of increased incidence of *A. ostoyae* root disease (Woods 1994; Simard and Heineman 1996;

Baleshta et al. 2005; Simard et al. 2005). Here, broadleaf trees appear to facilitate conifer survival through protection against root disease.

Mortality responses to weeding of broadleaf trees have varied with treatment method and intensity. While cut-stump glyphosate treatment has had no effect, mortality of Douglas-fir has increased 1.5–7 times following manual cutting and girdling treatments (Woods 1994; Simard and Heineman 1996; Baleshta et al. 2005; Simard et al. 2005). These mortality rates have the potential to reduce stocking levels in young stands and yield at rotation, even though mortality due to *A. ostoyae* may stabilize with time (Cruickshank et al. 1997). The difference in treatment method effect is likely associated with the competitiveness of *A. ostoyae* relative to other saprophytic fungi on manually stressed versus chemically killed paper birch roots. When birch is manually cut or girdled, the remaining stumps and roots are stressed, which stimulates infection by *A. ostoyae* and increases its potential to infect surrounding conifer hosts through rhizomorph spread or root contacts (Morrison et al. 2000, 2001). Increased growth of conifers released from competition with broadleaf trees have also been associated with increased infection rates (Rosso and Hansen 1998; Baleshta et al. 2005), and the faster Douglas-fir growth following girdling than manual cutting reported by Simard et al. (2005) corresponded to slightly higher mortality rates. Cut-stump glyphosate treatment, by contrast, has not increased mortality of Douglas-fir due to *A. ostoyae* (Simard and Heineman 1996; Simard et al. 2005), probably because the root systems are rapidly killed, thus providing a ready food source for saprophytes that are more aggressive than *A. ostoyae* (Morrison and Pellow 2002).

Where manual cutting of paper birch was used, mortality of Douglas-fir caused by *A. ostoyae* increased with cutting intensity (Baleshta et al. 2005). Any amount of birch cutting stimulated disease spread, but low-intensity cutting kept mortality rates relatively low. This may be explained by the greater inoculum load remaining on the more intensively cut sites, or by the faster growth of residual Douglas-fir. Vegetation-management treatments aimed at releasing conifers from competition with paper birch ought to balance growth gains against mortality losses in stands where *A. ostoyae* root disease is present. To achieve this balance on diseased sites, we suggest that paper birch be manually cut on a selective basis, targeting only individual Douglas-fir whose productivity is clearly compromised by competition.

Effects of weeding on plant community diversity

The broadcast weeding treatments in these studies reduced overstory broadleaf trees, affecting resource availability to the understory, but they did not expose mineral soil, and therefore had no effect on species richness or susceptibility to invasion of the plant community. However, according to Simard et al. (2005), structural diversity increased following cut-stump glyphosate and manual-cutting treatments, a result consistent with vegetation-management experiments carried out in other forest regions (Haeussler et al. 1999; Boateng et al. 2000). This resulted from removal of the broadleaf overstory, which was followed by increased

abundance of understory plant species and sprouting of cut paper birch stumps, which grew into shorter but more diverse structural layers. Here, the dominant broadleaf trees lost their controlling effects on the subordinate plant layers (Grime 1998). The mass-ratio theory of Grime (1998) predicts that dominant plants comprising most of the total plant biomass, such as broadleaf trees in our ecosystems, have an overwhelming influence on ecosystem function compared with subordinate or transient plants. The results of our study support this theory because Simard et al. (2005) and Baleshta et al. (2005) report that manual removal of broadleaf trees increased mortality due to *A. ostoyae* root disease. It is also predicted that removal of large paper birch trees will negatively affect cavity-nesting birds (Aitken et al. 2002) and songbirds (de Bellefeuille et al. 2001).

Future management pathways

Our review has shown that there are biological risks and considerable uncertainty associated with the expected outcomes of current management practices aimed at producing coniferous sawlogs in the mixed forest types of the ICH in British Columbia. Concerns about conifer regimes have been expressed by environmental groups, and the province seems likely to experience pressure for a more natural forestry (Hammond 1991), including mixed conifer and broadleaf management, to meet visual and biodiversity objectives, following trends in Germany and Scandinavia (Kenk 1992; Puttonen 1996) and elsewhere in Canada (Comeau et al. 2005). Less prominent are concerns about the economic wisdom of a conifer strategy. Broadleaf trees were once regarded as low-value species in Europe, in much the same way that they have been regarded in British Columbia in recent times. In Finland, prices for birch products rose rapidly in the 1980s (Puttonen 1996). Haight (1993) demonstrated the importance of stochastic price trends in his analysis of the economics of Douglas-fir and red alder (*Alnus rubra* Bong.) management in the Pacific North West. When investments have a long gestation period and prices are very uncertain, portfolio diversification is usually recommended.

What might be done to minimize biological, economic, and social risks in future forest management? In principle, we suggest implementation of a deliberate program of mixing conifer and broadleaf species both spatially and temporally across the landscape in much the same proportions in which they have occurred in recent history (Sachs et al. 1998). This would be preferable to the present approach, which we suggest is leading to forest simplification. For example, Simard et al. (2001) found that all ICH forest types in southern interior British Columbia were composed of a mixture of species prior to the mid-1990s, but since then, the current combined program of planting, brushing, and spacing has converted 39% of new forests to types composed predominantly of either Douglas-fir or lodgepole pine.

Our "mixtures" program would comprise three main pathways, to be applied in different proportions following harvesting across the landscape (Table 1). The first and most common pathway would involve enrichment of the naturally regenerated species mix with conifer species of current high commercial value, much as is done now, but at lower densities and thus lower establishment cost. Similar approaches

are used in German forests according to Kenk (1992). Stumping for root-disease abatement would be reduced and applied only in areas designated for high-yield management, with the remaining areas managed with resilient species mixtures that include broadleaf trees. Brushing costs would be reduced by removing the restrictions governing free-growing on the density and proximity of broadleaf species and allowing individual forest managers to apply selective treatments where crop-tree survival is threatened. Chemical brushing treatments would be applied on a limited basis to root disease infested sites as a way to reduce the risk of disease accentuation. Thus, managers might prescribe brushing to release western larch, which is very shade-intolerant, but not other species that are more tolerant of shade. The resulting stands would be managed as stratified mixtures following the precepts of Smith et al. (1997). Stand-improvement practices to control species composition or stem quality would be chosen on the basis of individual perceptions of species and stem value, market prices, and nonmarket values at the time, rather than making an irrevocable decision about future markets and values at the time of establishment. In the longer term, we forecast that such stands would be managed on a continuous-cover basis, with frequent entries, natural regeneration, and low additional management costs.

Methods for projecting the yield of mixed-species stands remain to be developed, even though vertical stratification of mixed ICH stands, and the respective height-growth patterns of species in mixtures, have been described (Cameron 1996). German experience with mixtures of beech and spruce suggests that results can be positive or negative depending upon the site (Kenk 1992). In Finland, experience with mixed stands of birch and pine or spruce suggests that total stand yield will be maximized with a birch component between 25% and 50% (Mielikainen 1996).

A second pathway would not involve enrichment planting with conifers, but instead rely upon natural regeneration. In the southern interior ICH forests, natural regeneration is abundant adjacent to retained forest patches within large patch cuts in most forest types, as well as in small harvested patches in the younger mixed forest types that originated from large wildfires 80–120 years ago (Vyse and DeLong 1994; Heineman et al. 2002). This pathway would be best suited to less productive sites and more remote locations, in keeping with the "semi-natural" approach recommended by Lieffers et al. (1996). These forests would receive minimal intervention, allowing natural stratification of the complex mixtures to occur.

The third pathway would apply to a much more limited portion of the landscape, where site productivity is highest. It would focus on achieving high productivity of individual species, or designed mixtures, over relatively short rotations. Conifer production would follow the standard prescription described earlier, but with more careful site selection, focusing on the most productive, mesic to subhygric portions of harvested sites. Emphasis would be placed on eliminating stand gaps and creating a sufficiently high density to permit future choices of high-quality crop trees. Plantations of birch are also a feasible option, and are common in Finland (Puttonen 1996), where production of genetically improved silver birch (*Betula pendula* Roth) reached 25 million seedlings annually in the 1990s. Birch should be established on

mesic to subhygric sites, where the species performs best (Simard and Vyse 1992; Simard et al. 2004a). Since genetically improved stock is not available at present, it should be established at high densities to allow stem selection in subsequent stand entries. Managers would be free to invest heavily in conifers and broadleaf species depending upon their perception of market and pest risks. This intensive-management element would have to be encouraged by government, since most forest lands in British Columbia are owned by the province. Private companies undertake forest regeneration as a condition of their harvesting tenure and not as an investment.

The management program we propose would be less costly in total, provide a greater range of forest products, provide a hedge against uncertainty in future wood-product markets, and provide opportunities for the development of nontraditional forest products, particularly those associated with broadleaf species, which are either limited or not available under the current management approach. It uses an ecological approach to management that integrates facilitative and competitive interactions in forest communities, and accounts for trade-offs between survival and growth that can result from manipulation of broadleaf trees. We suggest that this integration and understanding of interaction mechanisms and trade-offs are essential for predicting the net effect of vegetation-management treatments on forest ecosystem productivity and resilience. And while we cannot forecast with any confidence the effects of climate change, a mixed conifer-broadleaf forest should provide a greater buffering capacity than single-species forests against the stresses that it will impose, based upon our research and sound ecological theory.

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